MISSOURI SHRIMP FARM

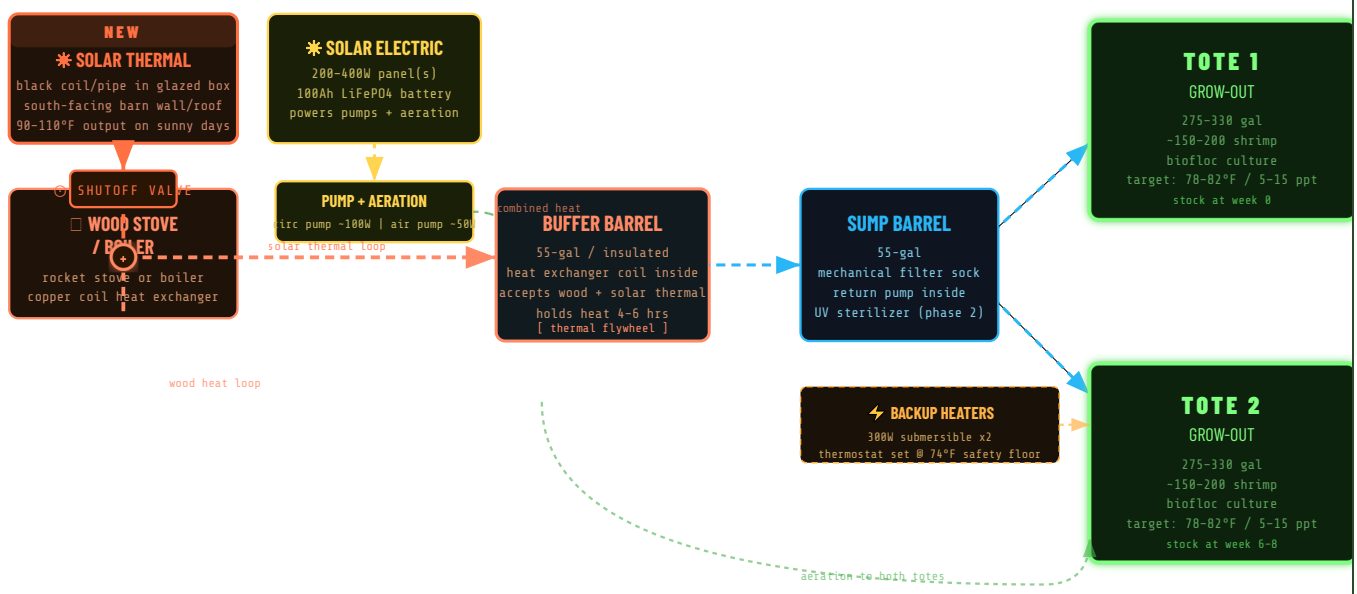
// TRIPLE-SOURCE HYBRID HEATING SYSTEM //

▲ V2 – SOLAR THERMAL COLLECTOR ADDED

SOLAR THERMAL (heat) SOLAR ELECTRIC (power) WOOD HEAT LOOP WATER CIRCULATION

AERATION BACKUP ELECTRIC HEAT

FULL SYSTEM FLOW – TRIPLE HYBRID



HEAT SOURCE A – NEW NEW ▲

SOLAR THERMAL

› Black painted coil/pipe in glazed box
› South-facing wall or roof mount
› Clear cover (old storm window works)
› ½" copper or PEX coil, 20-40ft

HEAT SOURCE B – $50-200 DIY

WOOD STOVE LOOP

› Rocket stove or outdoor wood boiler
› Copper coil heat exchanger in buffer
› Evenings, cloudy days, deep winter
› Free fuel – rural Missouri advantage

POWER SOURCE – ELEC $460-770

SOLAR ELECTRIC

› 200-400W panel(s) on barn roof
› 100Ah LiFePO4 battery bank
› 30A MPPT charge controller
› 1000W pure sine inverter
› Runs: pump, aeration, lights, monitors

› Shutoff valve — open when sunny
› Output: 90–110°F on good days
› Cost: near zero — you have materials!

› Needs tending 2–3x daily in winter
› Works independently of solar

› ~12–18hr runtime on full charge

THERMAL CORE ~$20 + foam

BUFFER BARREL

› 55-gal barrel, rigid foam insulation wrap
› Accepts heat from BOTH solar + wood
› Copper coil from each source inside
› Thermal flywheel — 4–6hr heat hold
› Add thermometer + low-temp alarm
› Most important component in system

FILTRATION + RETURN ~$20

SUMP BARREL

› 55-gal barrel, gravity fed from totes
› Mechanical filter sock at inlet
› Circulation pump pushes to totes
› Bio-media optional for nitrification
› UV sterilizer inline — add phase 2
› Clean filter sock weekly minimum

PRODUCTION — 2X GRO $50–150

IBC TOTES X2

› 275–330 gal each — ~600 gal total
› Stock Tote 1 at week 0
› Stock Tote 2 at week 6–8
› Rolling harvests every ~2 months
› 25–50 lbs per tote per harvest
› ~$10–20/lb at local markets

OPERATING MODES BY SEASON / WEATHER

☀ SUNNY DAY

Solar thermal valve: OPEN
Wood stove: OFF or minimal
Solar electric: charging battery
Best efficiency day — free heat all day

☁ CLOUDY DAY

Solar thermal valve: CLOSED
Wood stove: ON — primary heat
Solar electric: running on battery
Normal wood burn day

🌙 OVERNIGHT

Solar thermal: off (no sun)
Buffer barrel: coasting on stored heat
Backup heaters: armed at 74°F floor
Fire before bed = safe til morning

❄ DEEP WINTER

Solar thermal: supplemental when able
Wood stove: running hard, tend 3x/day
Backup heaters: may activate overnight
Insulate barn walls — priority investment

✱ SOLAR THERMAL BUILD TIPS

Paint your coil and backing board flat black (BBQ paint holds up). Use a south-facing angle of ~45–50° for Missouri latitude. Seal the glazing edges well — the greenhouse

⊙ SHUTOFF VALVE STRATEGY

Use a simple ball valve on the solar thermal outlet line. On sunny mornings, crack it open and feel the output temp — if it's warmer than your buffer barrel, open

💧 WATER CHEMISTRY TARGETS

pH: 7.5–8.3 | Ammonia: <1 ppm | Nitrite: <0.5 ppm | Salinity: 5–15 ppt | Temp: 78–82°F | DO: >5 mg/L. Temp swings >4°F in 24hrs will

effect inside the box is what makes it work even on cold sunny days. Even 30°F outside, inside the box can hit 120°F+.

it fully. If it's cooler (early morning, thin clouds), keep it closed so you don't accidentally cool the system down.

stress shrimp — your buffer barrel prevents this.

FULL BUILD COST ESTIMATE — TRIPLE HYBRID SYSTEM

Item	Est. Cost	Notes
2x IBC Totes	$50-150	Food-grade preferred
2x 55-gal Barrels	$40	Sump + buffer
Water Delivery	$100	Initial fill ~600 gal
PVC Plumbing + Fittings + Bulkheads	$100-200	Valves, pipe, unions
Circulation Pump	$40-80	Submersible or inline
Air Pump + Stones + Lines	$50-80	Heavy aeration = critical
Wood Stove / Rocket Stove	$50-200	DIY saves big here
Copper Coil — Wood Loop	$20-40	½" soft copper, ~20ft
☀ Solar Thermal Collector Frame/Glazing	$0-40	You have materials! Glazing = salvage window
☀ Copper/PEX Coil — Solar Thermal	$15-30	½" PEX is cheaper than copper
☀ Ball Valve (shutoff)	$5-12	½" brass ball valve
☀ Flat Black Paint (BBQ rated)	$8-15	Rust-Oleum high heat
Buffer Barrel Insulation (foam board)	$20-40	Wrap entire barrel
Backup Heaters x2 (300W) + Thermostat	$50-80	Safety floor @ 74°F
Salt Mix (~50 lbs for 600 gal @ 10ppt)	$50-80	Instant Ocean or similar
Test Kits — Full Suite	$50-80	API or Salifert brand
Post-Larvae Stock (PL12+, ~300-400)	$50-100	Order from reputable hatchery
PHASE 1 TOTAL	**$698-$1,227**	Full system incl. solar thermal
+ Solar Electric System (Phase 2)	$460-770	Panels, battery, inverter, controller
COMPLETE SYSTEM	**$1,158-$1,997**	**Full triple-hybrid, grid-independent**